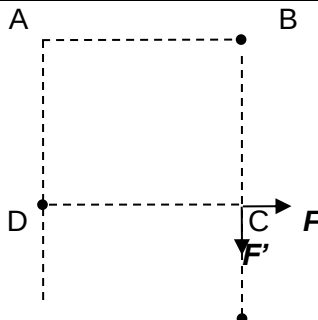


3	(a)	Electric potential at a point is the <u>work done per unit positive charge in bringing a charge from infinity to that point.</u>	B1
	(b)(i)	 Two arrows of equal length, correct direction with labels (i.e. Force on C by B, Force on C by D) – 1 Mark	B1
	(b)(ii)	$F = \frac{Q^2}{4\pi\epsilon_0 r^2} = (9 \times 10^9) \frac{(1.2 \times 10^{-6})^2}{(2.0 \times 10^{-2})^2} = 32.4 \text{ N}$ Resultant force = $\sqrt{(F^2 + F^2)} = F\sqrt{2} = 32.4\sqrt{2} = 45.8 \text{ N}$ Direction: Along A to C (ecf (i))	C1 A1 A1
	(b)(iii)	Distance from corner to centre, $r = \sqrt{2} \text{ cm}$ $V = \frac{Q}{4\pi\epsilon_0 r} = (9 \times 10^9) \frac{-1.2 \times 10^{-6}}{\sqrt{2} \times 10^{-2}} = -7.64 \times 10^5 \text{ V}$ Electric potential due to the three charges = $3V = 3(-7.64 \times 10^5)$ $= -2.29 \times 10^6 \text{ V}$	C1 C1 A1
	(b)(iv)	At 100m away, potential due to the 3 charges is negligible $W = q\Delta V = (1.2 \times 10^{-6})(-2.29 \times 10^6 - 0)$ $= -2.75 \text{ J}$	C1 A1

4	(a)	Any of 2 of the following:	
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